



Innovative Metabolomics Insights for Better Health

MWHW-24-0296-a University of Virginia 21 Bacterial Supernatant Samples Untargeted Plus Metabolomics -WOBI

Metware Biotechnology Inc.

www.metwarebio.com

Contents

1	Abstract	3
2	The experimental process	3
2.1	Sample information	4
2.2	Reagents and instruments	5
2.3	Sample extraction process	5
2.4	Data preprocessing	7
3	Data evaluation	7
3.1	Qualitative and quantitative analysis of metabolites	7
3.2	Quality control sample analysis	8
4	Reference	11

MWHW-24-0296-a_ University of Virginia 21 Bacterial Supernatant Samples Untargeted Plus Metabolomics -WOBI

1 Abstract

Metabolomics is the study of all metabolites and their dynamics in a biological system by performing qualitative and quantitative analyses. The data is often used to study the metabolic basis of observed phenotypes, to understand the response mechanisms under different physical, chemical, or pathological conditions, and to evaluate safety of food and drugs.

Untargeted metabolomics is a common approach to metabolomics research. The main idea is to perform qualitative and quantitative analysis, and identify statistically significant differential metabolites between different groups.

(1) For this project, 21 samples were selected and divided into 7 groups for metabolomics study.

Table 1: Number of identified metabolites

-	All	T3_positive	T3_negative
Number of metabolites identified	7911	4289	1979
Number of metabolites with MS2 structure	6180	3033	1760

Number of identified metabolites: Final_report/1.Data_Assess/metabolitesCount.xlsx

2 The experimental process

Ultra-performance liquid chromatography-tandem mass spectrometry (UPLC-MS/MS) is a technique used for accurate qualitative and quantitative analysis for various compounds. The main purpose of metabolomics analysis is to detect and identify metabolites with important biological significance by differentiate statistically significant differential metabolites between sample groups. The overall process is as follows:

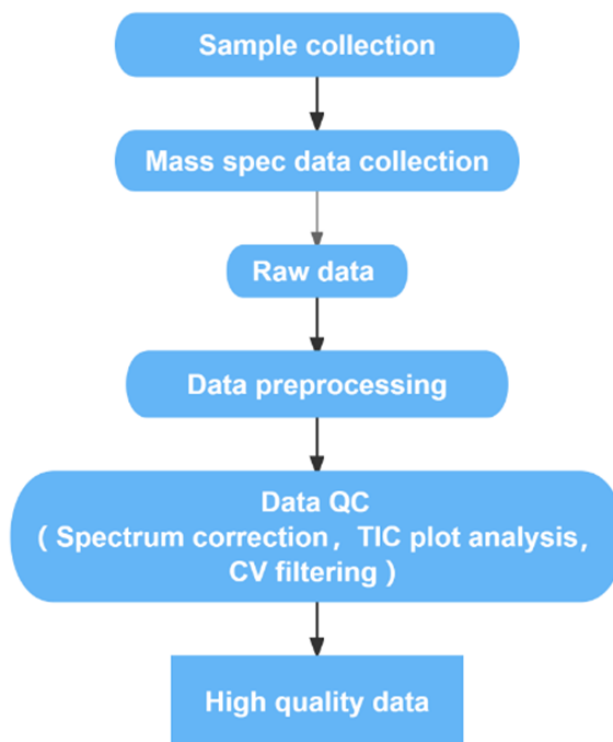


Fig 1: Flow chart of metabolomics analysis

2.1 Sample information

Each sample group and corresponding sample information are as follows:

Table 2: Sample information

Species	Tissue	Sample	Group
C. difficile	supernatant	1	Cd_R20291
C. difficile	supernatant	2	Cd_R20291
C. difficile	supernatant	3	Cd_R20291
B. producta	supernatant	4	B.producta
B. producta	supernatant	5	B.producta
B. producta	supernatant	6	B.producta
B. longum	supernatant	7	B.longum
B. longum	supernatant	8	B.longum
B. longum	supernatant	9	B.longum
N/A	media	10	BHI-blank

Table of sample information: Final report/1.Data_Assess/sample_info.xlsx

2.2 Reagents and instruments

Table 3: Information of standards and reagents

Compound	CAS	Purity	Brand
Methanol	67-56-1	more than 99.9%	Thermo Fisher
Acetonitrile	75-05-8	99.95%	Thermo Fisher
Acetic Acid	64-19-7	more than 99.7%	Thermo Fisher
Ammonium formate	540-69-2	more than 99.0%	Sigma
Ammonium hydroxide solution(25%)	1336-21-6	-	Sigma
Formic Acid	64-18-6	97.5-98.5%	Sigma
Standard	-	more than 98%	isoreag/TRC/TCI/Sigma

Table 4: Information of instruments

Name	Instrument	Brand	Address
Mass Spectrometer	TripleTOF 6600+	SCIEX	California, USA
UHPLC	ExionLC AD	SCIEX	California, USA
Centrifuge	5430R	Eppendorf	Hamburg, Germany
Vortex mixer	MI0101002	HFour E's	Guangzhou, China
Electronic balance(1/100000)	AS60/220.R2 Plus	Radwag	Poland
Centrifugal concentrator	CentriVap	LABCONCO	Missouri Kansas, USA
Ultrasonic cleaner	CD-F15	Olenyer	China
Pipette	Research plus	Eppendorf	Hamburg, Germany
Automatic workstation	Biomek i5	Beckman Coulter	California, USA
Uniform-Seal Heat Sealer	FM 5200	foodsaver	California, USA

2.3 Sample extraction process

2.3.1 Liquid samples class II

Samples stored at -80 °C was thawed on ice and vortexed for 10 s. 50 ul of the sample and 150 µL of the extraction solution (can : Methanol = 1:4, V/V) containing internal standard was mixed in a 2 mL microcentrifuge tube. The sample was vortex for 3 min and centrifuged at 12000 rpm for 10 min (4 °C). 150 µL of the supernatant was collected and placed in -20 °C for 30 min followed by centrifugation at 12000 rpm for 3 min (4 °C). A 120 µL aliquot of the supernatant was used for LC-MS analysis.

2.3.2 Liquid phase conditions (T3)

Chromatographic column: Waters ACQUITY UPLC HSS T3 1.8 μ m, 2.1 mm * 100 mm

Mobile phase A: ultrapure water (0.1 % formic acid added)

Mobile phase B: acetonitrile (0.1 % formic acid added)

Column temperature: 40 °C; Flow rate: 0.40 mL/min; Injection volume: 4 μ L

Table 5: Elution gradient for T3

Time (min)	A (%)	B (%)
0.0	95	5
2.0	80	20
5.0	40	60
6.0	1	99
7.5	1	99
7.6	95	5
10.0	95	5

2.3.3 Liquid phase conditions (HILIC)

Chromatographic column: ACQUITY Premier BEH Amide 1.7 μ m, 2.1 mm * 150 mm

Mobile phase A: 60 % ACN, 30 % water, and 10 % MeOH with 20 mM ammonium formate, pH 10.6

Mobile phase B: 40 % ACN and 60 % water with 20 mM ammonium formate, pH 10.6

Column temperature: 40 °C; Flow rate: 0.40 mL/min; Injection volume: 4 μ L

Table 6: Elution gradient for HILIC

Time (min)	A (%)	B (%)
0.00	95	5
3.50	70	30
5.50	5	95
6.50	5	95
6.51	95	5
10.00	95	5

2.3.4 Mass spectrum conditions

Table 7: Mass spectrum conditions for AB Sciex TripleTOF 6600

Parameter	ESI+	ESI-
Duration (min)	10	10
IonSpray Voltage (V)	5000	-4000
Temperature (°C)	550	450
Ion Source Gas1 (psi)	50	50
Ion Source Gas2 (psi)	60	60
Curtain Gas (psi)	35	35
Declustering Potential (V)	60	-60
MS1 Collision Energy (V)	10	-10
MS2 Collision Energy (V)	30	-30
Collision Energy Spread (V)	15	15

2.4 Data preprocessing

The original data file acquired by LC-MS was converted to mzML format by ProteoWizard. Peak extraction, peak alignment and retention time correction were performed by XCMS program. The peaks with missing rate >50% in each group of samples were filtered. The blank values were filled with KNN, and the peak area was corrected by SVR method. The metabolites were annotated by searching the MetwareBio's in-house database, integrated public database, prediction database and metDNA. Finally, substances with a comprehensive identification score above 0.5 and a CV value of QC samples less than 0.3 were extracted, and then positive and negative mode were combined (substances with the highest qualitative grade and the lowest CV value were retained) to obtain the all_sample_data.xlsx file.

3 Data evaluation

3.1 Qualitative and quantitative analysis of metabolites

The metabolite ID, relative content and corresponding metabolite names of some metabolites detected in this experiment are shown in the following table:

Table 8: Information of metabolite detected in sample

Index	1	2
MW0169982	134348.22	226560.90
FDATN01466	232298.36	336790.58
MEDP1681	304264.46	301667.78
MW0169547	391908.05	516325.27
MW0168692	5494369.55	6083681.77
MW0107537	248809.79	287256.04
MW0169283	8387.36	11500.51
MW0106559	1196517.94	1556903.81
MEDP0545	28466.40	14112.83
MEDN0305	70550.25	75659.68

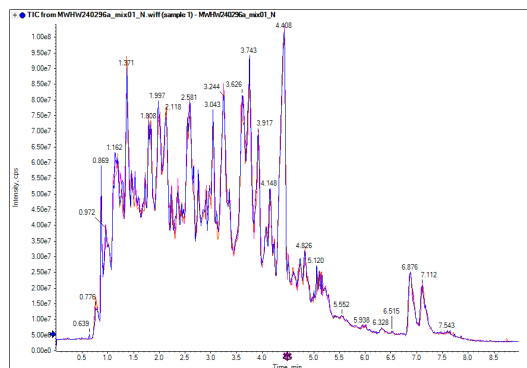
Information of metabolite detected in sample: Final report/1.Data_Assess/all_sample_data.xlsx

3.2 Quality control sample analysis

3.2.1 Total ion flow chromatogram

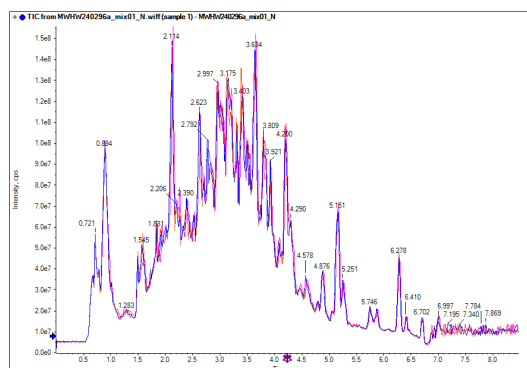
A quality control (QC) sample was prepared from a mixture of all sample extracts to examine the reproducibility of the entire metabolomics process. During data collection, one quality control sample was generally inserted for every 10 test samples.

Reproducibility of metabolite extraction and detection process was assessed by analyzing overlapping total ion flow diagram (TIC diagram) from different QC samples. High overlapping rate of TIC diagrams indicates high stability of the instruments throughout the data acquisition process



10/20/2024 1:31:28 PM

(a) MWHW-24-0296-a_HILIC_QC_MS_tic_overlap



10/20/2024 1:25:39 PM

(b) MWHW-24-0296-a_neg_QC_MS_tic_overlap

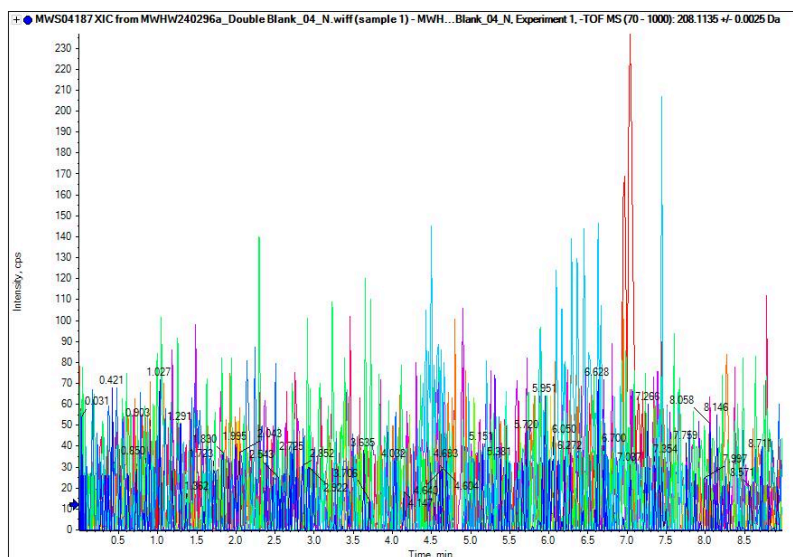
Fig 2: TIC overlap diagram detected by QC sample essence spectrum

Note: Superimposed spectrum from different QC samples. The results showed that the spectrum of total ion flow were highly consistent indicating that the signal stability was good when the same sample was detected at different times by mass spectrometry. N stands for negative ion mode and P stands for positive ion mode.

TIC overlap diagram detected by QC sample essence spectrum: Final report/1.Data_Assess/*/QC/
QC_MS_tic_overlap.*

3.2.2 Peak appearance of internal standards in blank samples

Blank samples were interspersed throughout the experiment, and their peaks can reflect whether there are compound residues from the detection process. The figure below shows that no obvious internal standard peaks were detected in the blank samples, indicating that possibility of cross-contamination between the samples is minimal.



10/3/2024 1:32:23 PM

Fig 3: EIC diagram of internal label in blank sample

Note: The signals in the EIC diagram are noise peaks, and the internal standard substance has no obvious signal peak at the corresponding time.

EIC diagram of internal label in blank sample: Final report/1.Data_Assess/*/QC/*BLANK_EIC.png

3.2.3 Stability of internal standards in QC samples

Internal standards with known concentrations were added to the QC samples for assessing variations between samples. The smaller the variation, the more stable the detection process and the higher the data quality.

Table 9: Stability of internal standard in QC samples

Index	Q1 (Da)	RT (min)	CV
MWS20853-IS-HN	281.1114	2.990000	0.0296
MWS04113-IS-HN	137.1313	1.910000	0.0516
MWS0094-IS-HN	121.0444	3.673333	0.0588
MWS04218-IS-HN	244.0899	1.900000	0.0742

Stability of internal standard in QC samples: Final report/1.Data_Assess/*/QC/*internal_standard_information.x

4 Reference

1. L. Eriksson, E.J., N. Kettaneh-Wold, J. Trygg, C. Wikström, and S. Wold, Multi- and Megavariate Data Analysis Part I Basic Principles and Applications, Second edition Umetrics Academy: Sweden, 2006. https://www.researchgate.net/publication/285755118_Multi-_and_Megavariate_Data_Analysis_Part_I_Basic_Principles_and_Applications_Second_revised_and_enlarged_edition
2. Chen, Y., et al., RRLC-MS/MS-based metabolomics combined with in-depth analysis of metabolic correlation network: finding potential biomarkers for breast cancer. Analyst, 2009.134(10): p. 2003-11. <http://dx.doi.org/10.1039/b907243h>
3. Viant M R, Kurland I J, Jones M R, et al. How close are we to complete annotation of metabolomes?[J]. Current opinion in chemical biology, 2017, 36: 64-69. <https://www.sciencedirect.com/science/article/pii/S1367593117300054>
4. Liang L, Rasmussen M L H, Piening B, et al. Metabolic dynamics and prediction of gestational age and time to delivery in pregnant women[J]. Cell, 2020, 181(7): 1680-1692. e15. <https://www.sciencedirect.com/science/article/pii/S009286742030564X>